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Sem : 7

Div : A

Roll no : 19

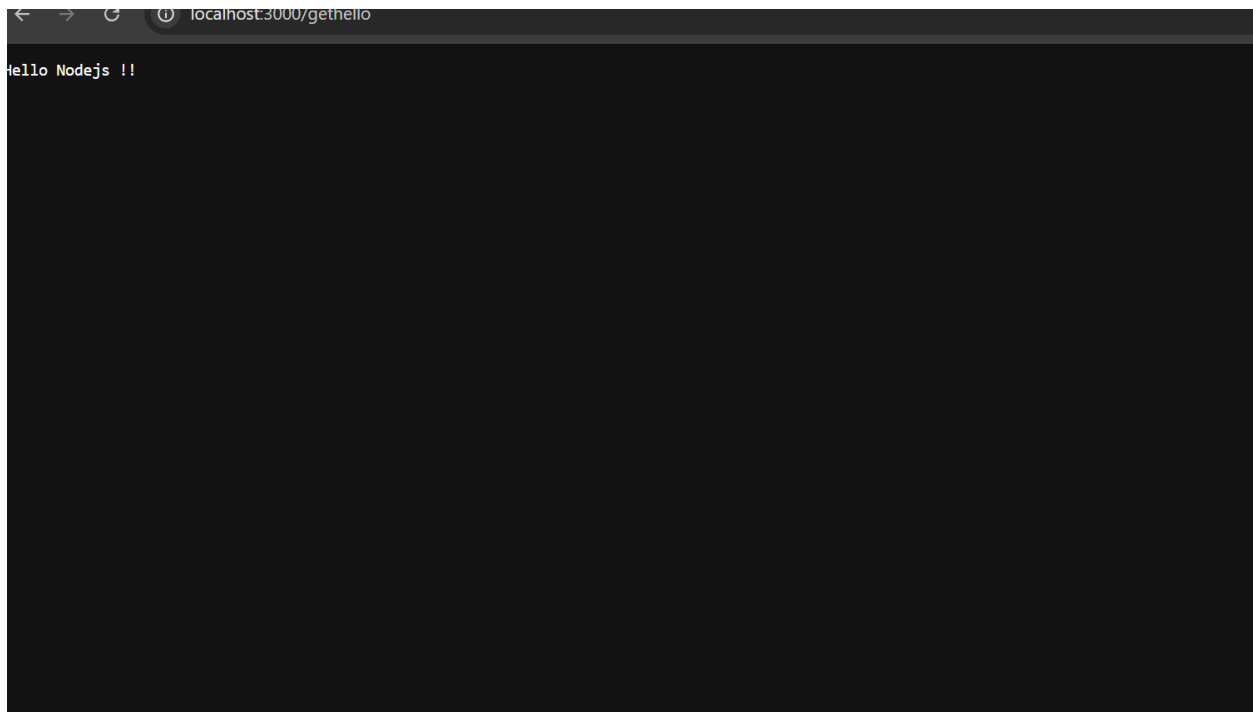
Subject : Application Development Using Full

Stack (701)

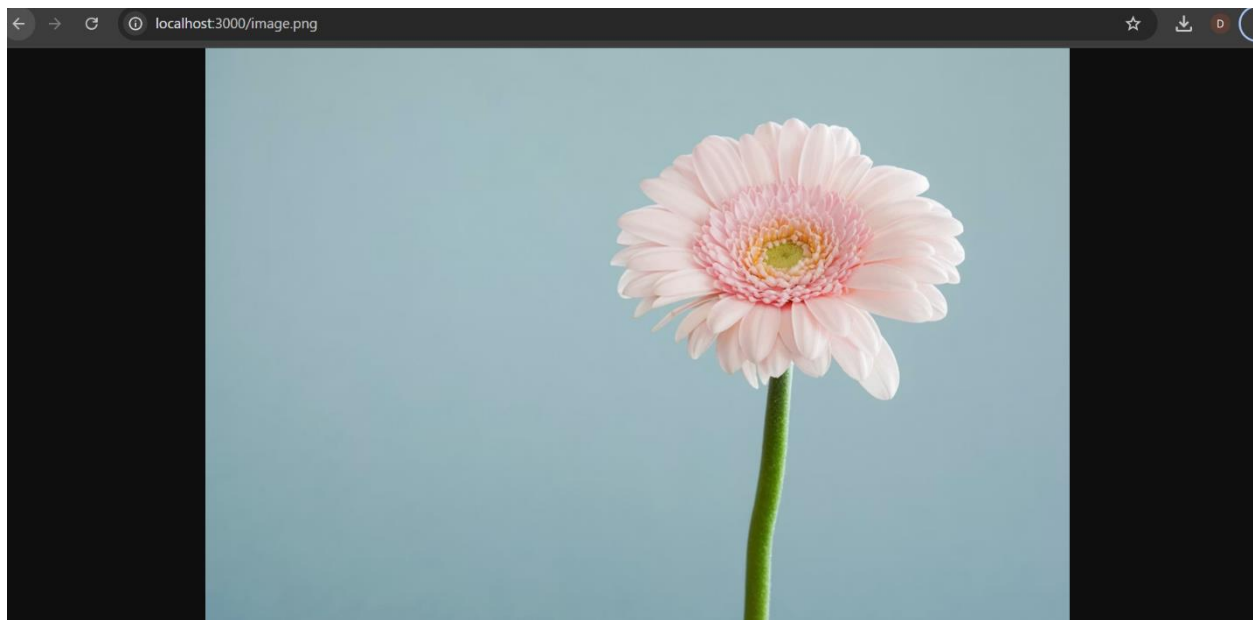
Practical_Assignmnet : 1

Question 1 :

Develop nodejs application with following requirements: - Develop a route `"/gethello"` with GET method. It displays `"Hello NodeJS!!"` as response. - Make an HTML page and display. - Call `"/gethello"` route from HTML page using AJAX call. (Any frontend AJAX call API can be used.)



Question 2 :Develop a web server which serves static resources.



Question 3 :

Develop a module for domain specific chatbot and use it in a command line application.

A screenshot of the Visual Studio Code editor. The Explorer sidebar on the left shows a project structure with folders 'Q1' through 'Q12' and files 'package-lock.json', 'package.json', 'static_server.js', 'app.js', and 'chatbot.js'. The main editor area shows the 'app.js' file with the following code:

```
1 var Chatbot = require('./Chatbot');
2 var readline = require('readline');
3 var rl = readline.createInterface(process.stdin, process.stdout);
4 rl.setPrompt("You ==> ");
5 rl.prompt();
6 rl.on('line', function (message) {
7     if (message == "exit") {
8         console.log("Bot ==> Goodbye!");
9         rl.close();
10    }
11    else {
12        console.log("Bot ==> " + Chatbot.ChatbotReply(message));
13        rl.prompt();
14    }
15 });
```

The bottom panel shows the 'TERMINAL' with the following output:

```
PS C:\MscIt sem-7\node\assignment_1\Q3> node app
You ==> hi
Bot ==> Hi!
You ==> how are you
Bot ==> I'm fine
You ==> age
Bot ==> I'm 25
You ==> where live you
Bot ==> I live in India
You ==> hobby
Bot ==> Sorry, I didn't get it :(
You ==> exit
Bot ==> Goodbye!
Chat ended.
```

Question 4 :

Write a program to create a compressed zip file for a folder.

```
JS q4.js  ●  <> sample.html  JS file_extract.js  JS q5.js  JS q7.js  ●  JS q8.js
que4 > JS q4.js > ...
 2  const fs = require("fs");
 3  const zlib = require("zlib");
 4  const path = require("path");
 5  const files = fs.readdirSync("./sample");
 6
 7  files.forEach((file) => {
 8      const inputPath = path.join("./sample", file);
 9
10      if (fs.statSync(inputPath).isFile()) {
11          const input = fs.createReadStream(inputPath);
12
13          const output = fs.createWriteStream(
14              path.join("./sample", file + ".gz")
15          );
16
17          input
18              .pipe(zlib.createGzip())
19              .pipe(output)
20              .on("finish", () => {
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
C:\node\que4>node q4
image.png compressed successfully

C:\node\que4>node q4
image.png.gz compressed successfully
sample.html compressed successfully
image.png compressed successfully
```

Question 5 :

Write a program to extract a zip file.

```
JS q4.js  ●  <> sample.html  JS file_extract.js  JS q5.js  JS q7.js  ●  JS q8.js
que4 > JS q4.js > ...
 2  const fs = require("fs");
 3  const zlib = require("zlib");
 4  const path = require("path");
 5  const files = fs.readdirSync("./sample");
 6
 7  files.forEach((file) => {
 8      const inputPath = path.join("./sample", file);
 9
10      if (fs.statSync(inputPath).isFile()) {
11          const input = fs.createReadStream(inputPath);
12
13          const output = fs.createWriteStream(
14              path.join("./sample", file + ".gz")
15          );
16
17          input
18              .pipe(zlib.createGzip())
19              .pipe(output)
20              .on("finish", () => {
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
C:\node\que4>node q4
image.png compressed successfully

C:\node\que4>node q4
image.png.gz compressed successfully
sample.html compressed successfully
image.png compressed successfully
```

Question 6 :

Write a program to promisify fs.unlink function and call it.

```
que6 > JS q6.js > ...
2
3   const fs = require("fs");
4   const util = require("util");
5   const unlink = util.promisify(fs.unlink);
6
7   unlink("sample.txt")
8     .then(() => {
9       console.log("File deleted successfully");
10    })
11    .catch((err) => {
12      console.log("Error:", err.message);
13    });
14
15
```

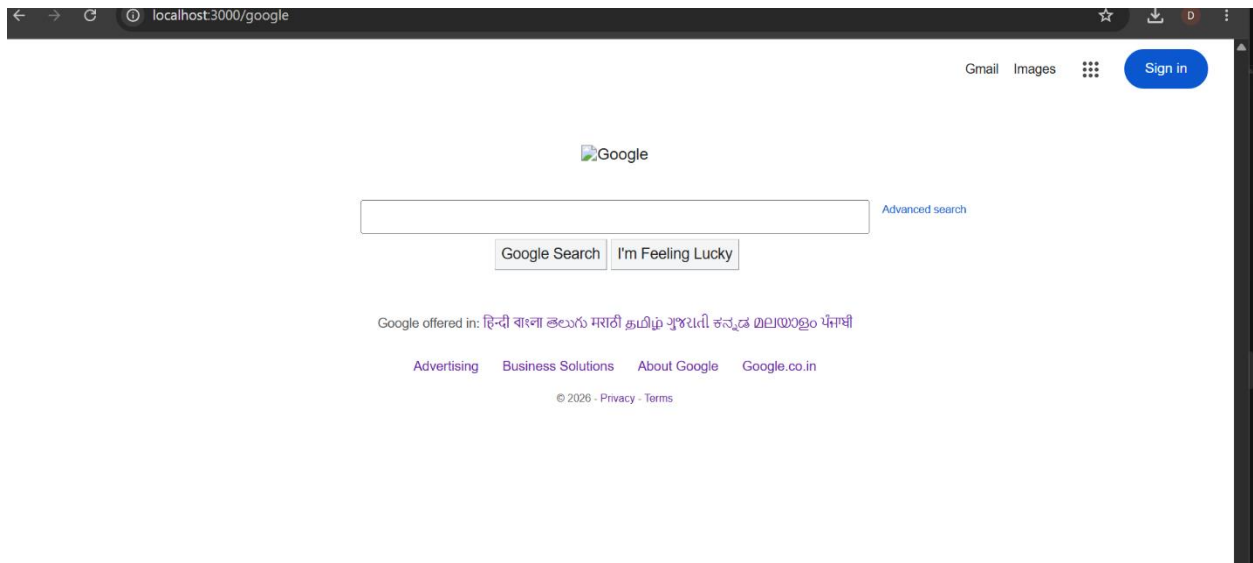
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
C:\node\que6>node q6
File deleted successfully

C:\node\que6>cd..
```

Question 7 :

Develop a node web server with a route /google which can fetch data of google page using fetch() using async-await model and show as a response.



Question 8 :

Set a server script, a test script and 3 user defined scripts in package.json file in your nodejs

```
1
2 {
3   "name": "question8",
4   "version": "1.0.0",
5   "description": "Node.js Practical",
6   "main": "build.js",
7   "type": "commonjs",
8   "scripts": {
9     "server": "node build.js",
10    "test": "echo Testing...",
11    "start": "node build.js",
12    "dev": "node build.js",
13    "hello": "echo Hello Node"
14  },
15  "author": "",
16  "license": "ISC"
17 }
18
19
20
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
C:\node\que8>npm test

> question8@1.0.0 test
> echo Testing...
```

Question 9 :

A program which calls useful functions in fs module.

que9 > JS q9.js > ...

```
22 console.log("Size:", info.size, "bytes");
23 console.log("File:", info.isFile());
24 console.log("Directory:", info.isDirectory());
25
26 fs.renameSync("sample.txt", "sample2.txt");
27 console.log("\n....File renamed succefully....");
28
29 fs.copyFileSync("sample2.txt", "sampleC.txt");
30 console.log("\n....File copied successfully....");
31
32 fs.unlinkSync("sample2.txt");
33 console.log("\n....File Deleted successfully.");
34
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

C:\node\que9>node q9

....File created successfully....

...Data appended successfully....

File reading succesfullly

Hello Node.js

I am appending data to the file

....file exists....

Size: 46 bytes

File: true

Directory: false

....File renamed succefully....

Question 10 :

A program which uses global objects in nodejs. Also read command line arguments and print it.


```

que10 > JS q10.js > ...
12 console.log("\nArguments :");
13
14 const args = process.argv.slice(2);
15
16 if (args.length === 0) {
17     console.log("No arguments provided.");
18 } else {
19     args.forEach((arg, index) => {
20         console.log(`Argument ${index + 1}: ${arg}`);
21     });
22 }
23

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

C:\node>cd que10

C:\node\que10>node q10

Current file path : C:\node\que10\q10.js

Current directory : C:\node\que10

Version : v24.18.0

Operating system : win32

Arguments :

No arguments provided.

Question 11 :

Develop a useful package and publish it on npmjs.com. Write a program to use it.

Question 12 :

. Write a node program having at least 3 dependent tasks and call using chained promises.

```
30     .then((result) => {
31         console.log(result);
32         return payFees(result);
33     })
34     .then((result) => {
35         console.log(result);
36         return genReceipt(result);
37     })
38     .then((result) => {
39         console.log(result);
40     })
41     .catch((err) => {
42         console.log("Error:", err);
43     });
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

C:\node>cd que12

C:\node\que12>node q12

Task 1: Student Registered

...Registration Completed...

Task 2: Fees Paid

....Fee Payment Completed....

Task 3: Fee receipt generate

....Fees Status updated....

C:\node\que12>